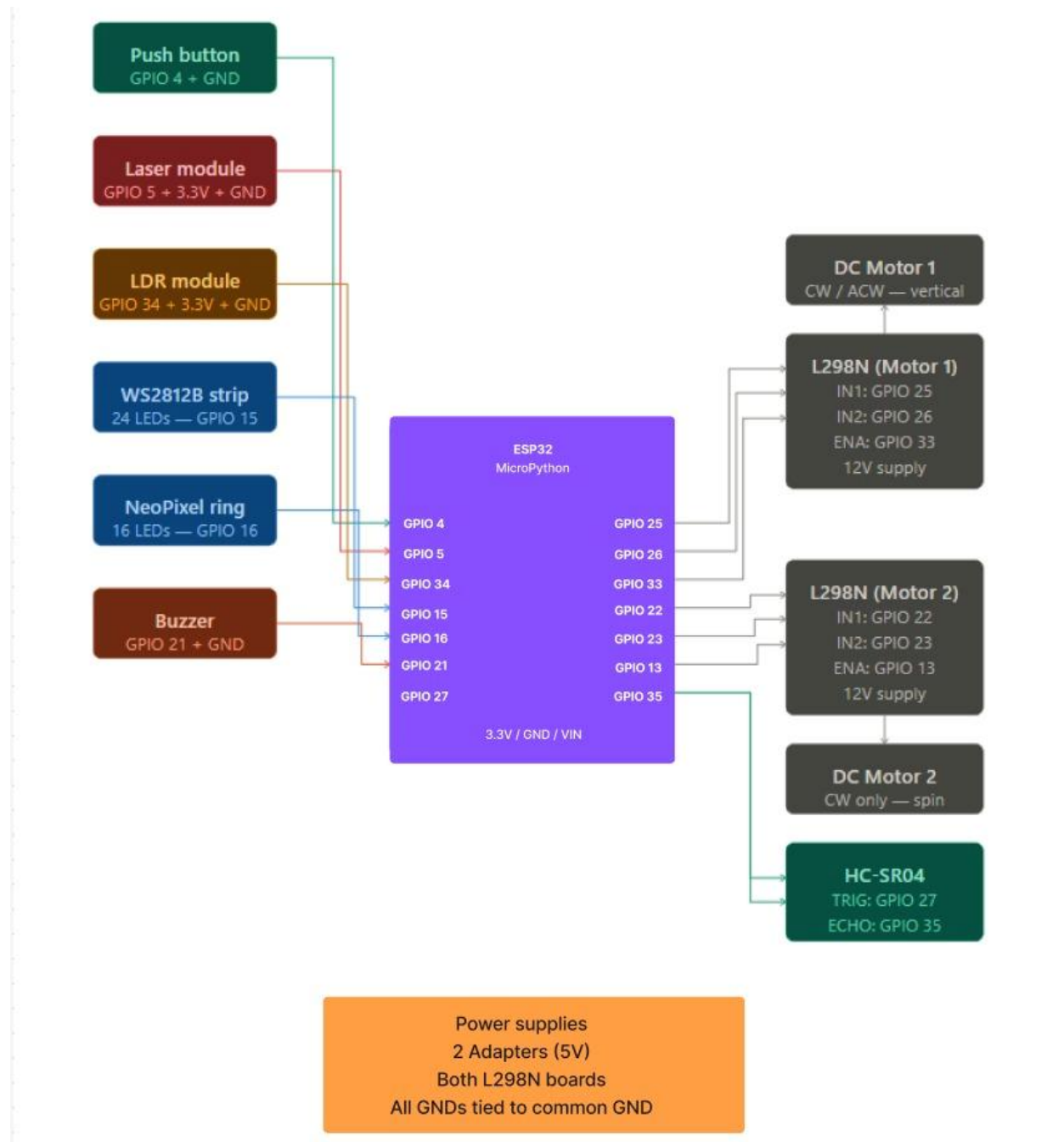


Electronics Documentation

1. Electronics Block Diagram



2. Components Table

Component	Quantity	Purpose	Justification
Push Button	1	The button turns the laser module on	Enables user-triggered activation, initiating the system sequence.
Laser Module	1	To have a dynamic start to the ride using mirrors in the style of a maze	Emits a focused beam suitable for precise path alignment across multiple reflections.
Light Dependent Sensor	1	To detect the laser's light for the maze to complete	Detects successful beam arrival to confirm maze completion and has adjustable sensitivity.
Neopixel	1	Blinks to signal the ride is ready to play with. Also for cool light show	Integrates individually addressable LEDs, enabling programmable, high-resolution visual feedback at once
LEDs and Buzzers	Multiple	For signalling systems are ready and for extra effects	Deliver low-power, real-time visual and auditory outputs for clear system state indication.
DC Motor	1	To spin the gear of the linear actuator	Supplies continuous rotational motion with sufficient torque for driving mechanical components like linear actuators.
Stepper Motor	1	To spin the middle rod that rotates swing	Allows discrete, step-controlled rotation for accurate positional movement of the central rod.

Ultrasonic Sensor	1	To detect the hand movement pattern	Utilizes distance measurement to detect and interpret proximity-based hand gestures without physical contact.
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